CURRICULUM VITAE

# Saeed Rafieyan

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## EDUCATION

2026 – present

### PhD

*Institute of Physical Chemistry, Polish Academy of Sciences (IChF PAN)* · Warsaw, Poland

Thesis: *Inverse design-assisted biodegradable modular dressings for negative pressure wound healing*

- Digital Manufacturing of Biomimetic Systems group.

2018 – 2022

### MSc in Chemical Engineering – Biomedical Engineering

*Tarbiat Modares University* · Tehran, Iran

Thesis: *Predicting cell behavior on cardiac tissue engineering scaffolds using machine learning algorithms*

- GPA: 3.78 / 4

2013 – 2017

### BSc in Chemical Engineering

*Tafresh University* · Tafresh, Iran

Thesis: *Simulation of the biodiesel production process with Aspen HYSYS*

## RESEARCH EXPERIENCE

2025 – present

### Research assistant – data scientist

*Jiangnan University, Dr. Madadi's group* · Wuxi, China

- Machine-learning and deep-learning models for biofuel production and process optimisation, resulting in peer-reviewed publications.

2023 – 2025

### Research assistant – data scientist

*Tarbiat Modares University, Dr. Fakhroleslam's group* · Tehran, Iran

- Machine-learning and deep-learning models for distillation and separation processes; chemical-engineering text mining.

2019 – 2025

### Research assistant – data scientist

*Tarbiat Modares University, Prof. Vasheghani-Farahani's group* · Tehran, Iran

- Developed and led AI-driven research projects, including MLATE (three versions, web app at mlate.ir) and TissueGPT, a language model fine-tuned on tissue-engineering literature.

2022 – 2024

### Data scientist

*Men's Health & Reproductive Health Research Centre, Shohadaye Tajrish Hospital* · Tehran, Iran

- Multimodal deep learning for semen analysis and drug-effectiveness prediction from clinical data and imaging.

## INDUSTRY EXPERIENCE

2023 – 2025

### Data scientist

*HiWEB* · Tehran, Iran

- Churn, next-purchase and purchase-value models for marketing; ASR and LLM-based quality control for Persian call-centre audio; dashboards in Flask and Plotly Dash.

2021

### Data scientist

*YecomSoft* · Tehran, Iran

- Persian NLP and text-to-speech for audiobook production.

2020

### Django web developer (intern)

*MAPSA* · Tehran, Iran

- Backend and REST APIs for an online retail platform.

## PUBLICATIONS

[9] **Bioprinting-Enabled Customization of Biohybrid Actuators: Scaling Dimensions and Functional Complexity**

Pajouhande M, Celikkin N, **Rafieyan S**, et al.

*Gels* · 2026 · `doi:10.3390/gels12100923`

[8] **MLATE V3: An Open-Source Multi-Tissue AI Framework for Data-Driven Optimization of 3D-Printed and Bioprinted Scaffolds**

**Rafieyan S**, Partovi Nasr M, Ansari E, Kheradvar Kolour A, Banimohamad-Shotorbani B, Vasheghani-Farahani E

*Biofabrication* · 2026 · `doi:10.1088/1758-5090/aeb162` IN PRESS

[7] **Data-driven insights for enhanced cellulose conversion to 5-hydroxymethylfurfural using machine learning**

Qiao Y, Kargaran E, Ji H, Madadi M, **Rafieyan S**, Liu D

*Bioresource Technology* · 2025 · doi:10.1016/j.biortech.2025.132582

[6] **A practical machine learning approach for predicting the quality of 3D (bio)printed scaffolds**

**Rafieyan S**, Ansari E, Vasheghani-Farahani E

*Biofabrication* · 2024 · doi:10.1088/1758–5090/ad6374

[5] **Acetone-butanol-ethanol fermentation products recovery: Challenges and opportunities**

**Rafieyan S**, Boojari MA, Setayeshnia A, et al.

*Chemical Engineering Research and Design* · 2024 · doi:10.1016/j.cherd.2024.04.021

[4] **MLATE: Machine learning for predicting cell behavior on cardiac tissue engineering scaffolds**

**Rafieyan S**, Vasheghani-Farahani E, Baheiraei N, Keshavarz H

*Computers in Biology and Medicine* · 2023 · doi:10.1016/j.compbiomed.2023.106804

[3] **A Review of Recent Advances in Natural Polymer-Based Scaffolds for Musculoskeletal Tissue Engineering**

Fan J, Abedi-Dorcheh K, Sadat Vaziri A, **Rafieyan S**, et al.

*Polymers* · 2022 · doi:10.3390/polym14102097

[2] **Predicting cell behavior on cardiac tissue engineering scaffolds using machine learning algorithms**

**Rafieyan S**

*MSc thesis, Tarbiat Modares University, Tehran* · 2022

[1] **Process Simulation of Biodiesel Production from Soybean Oil with Aspen HYSYS: A Comparative Study of two different processes**

**Rafieyan S**, Bayat A

*11th International Chemical Engineering Congress & Exhibition (IChEC 2020), Fouman, Iran* · 2020

## SOFTWARE AND DATA

**MLATE** – Machine Learning Applications in Tissue Engineering: open models and data that predict how a scaffold will print and how cells will respond to it, rank candidate formulations, and suggest what to fabricate next.

**TissueGPT** (doi:10.57967/hf/8917) — A BioGPT language model fine-tuned on tissue-engineering literature, for summarising papers, generating domain text and supporting scientific writing.

**ECM Program Atlas** — A reproducible, Matrisome-based framework for mapping extracellular-matrix gene programs across human tissues, validated on external transcriptomic and protein-level data, to support ECM-informed biomaterial design.

## SERVICE

Peer reviewer for IOP Publishing journals (IOP Trusted Reviewer), including Physics in Medicine and Biology, Biomedical Materials, Biomedical Physics & Engineering Express, Engineering Research Express and Progress in Biomedical Engineering; also Results in Surfaces and Interfaces (Elsevier).

## SKILLS

### Machine learning

Regression, classification, clustering, feature engineering, model selection, Bayesian optimisation, gradient boosting (XGBoost, LightGBM, CatBoost), scikit-learn

### Deep learning and NLP

PyTorch, TensorFlow, Hugging Face Transformers, CNNs, RNNs, multimodal learning, LLM fine-tuning, text mining, topic modelling

### Software and data

Python, NumPy, pandas, Matplotlib, Plotly, Dash, Streamlit, Flask, FastAPI, Docker, GitHub Actions, PostgreSQL, MySQL, SQLite

### Laboratory

Cell culture, MTT assay, decellularisation, hydrogel preparation, scaffold fabrication, 3D printing, freeze-drying

## LANGUAGES

**English** · Fluent (TOEFL iBT 90)

**Persian** · Native

## ADDITIONAL TRAINING

Neural Networks and Deep Learning · Coursera · 2020

Deep Learning with PyTorch: Image Segmentation · Coursera · 2022

Deep Learning with PyTorch: Object Localization · Coursera · 2022

Data Visualization using Plotly · Coursera · 2022

Introduction to Genomic Technologies · Coursera · 2022

Python for Genomic Data Science · Coursera · 2022

## REFERENCES

Available on request.